



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

**Salem Terminal TO NAMAKKAL TAL. RIG
URIMAIYALARGAL SA**

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

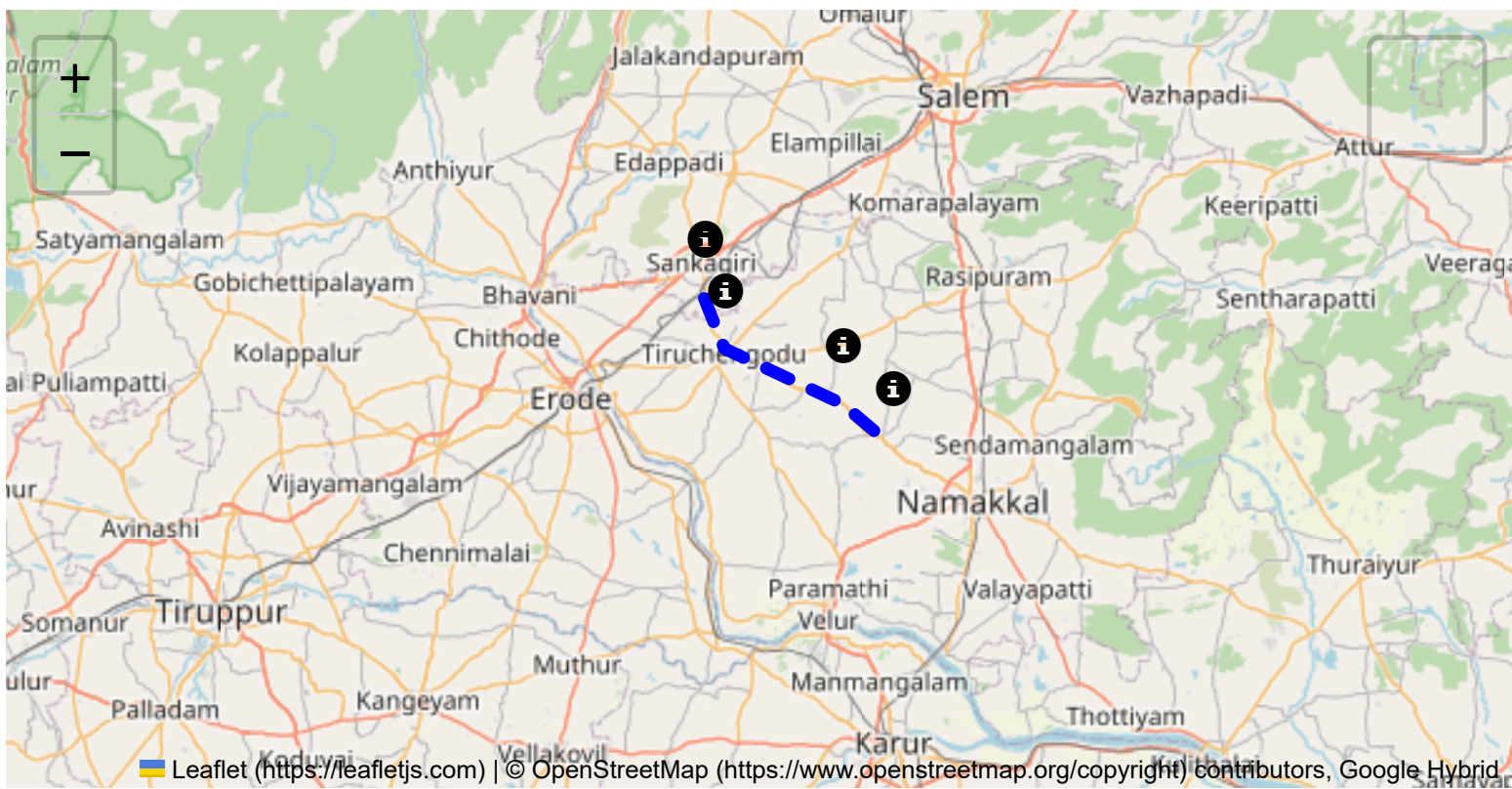
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 32.34 km
Estimated Duration: 0.7 hours
Adjusted Duration (Heavy Vehicle): 0.9 hours
Start: (11.4381, 77.8734)
End: (11.276951, 78.080055)



Welcome to the Journey Risk Management Study

Certainly! Here is a comprehensive analysis of the route from Sangagiri to Akkalampatti, with a focus on potential hazards and considerations for a truck driver:

1. Overview of the Route Map:

- The route covers approximately 32.34 kilometers and passes through key areas like Puthur, Tiruchengode, and Koothampoondi before reaching Akkalampatti. It involves a combination of

local roads and state highways, presenting a mix of rural and semi-urban driving conditions.

2. Typical Weather Conditions and Potential Hazards:

- Tamil Nadu generally experiences tropical weather with hot summers (March to June) and a monsoon season (June to September). Heavy rains during monsoon can lead to slick road conditions and occasional flooding, particularly affecting certain low-lying rural areas.

3. Traffic Patterns and Congestion-Prone Areas:

- The route could experience heavy traffic around Tiruchengode, especially near bus stands and markets. Peak congestion is likely during morning and evening rush hours (8-10 AM and 5-7 PM). Be cautious around school zones and traditional market days when foot traffic might increase.

4. Assessment of Road Quality and Infrastructure:

- As you traverse from Sangagiri to Akkalampatti, expect varying road conditions. Highways generally have better maintenance compared to local roads, which may have potholes or uneven surfaces. Watch for narrow stretches, particularly on village roads.

5. Suggestions for Alternative Routes for Emergencies:

- In case of roadblocks or severe congestion, consider using alternate routes like the Namakkal to Salem bypass roads, although they may extend travel time, they could offer less traffic and better road conditions.

6. Local Regulations on Hazardous Material Transport:

- Ensure compliance with local Tamil Nadu transport laws, which may include restricted times for transporting hazardous materials to reduce traffic risk. Check for any particular permits required for specific materials.

7. Historical Incidents Involving Heavy Vehicles:

- While specific data is less available, Tamil Nadu has witnessed incidents involving heavy vehicles due to negligence or overload. Drive cautiously through towns and adhere to speed limits to mitigate accident risks.

8. Environmental Considerations and Sensitive Areas:

- Be aware of environmentally sensitive areas such as forest patches or water bodies. Minimize disturbances and ensure no leak of hazardous materials, particularly near agricultural lands and villages.

9. Analysis of Communication Coverage:

- Generally, there is good network coverage throughout the route, but intermittent dead zones may exist in rural patches. Ensure communication devices are fully charged and pre-plan breaks in areas with reliable connectivity.

10. Estimated Emergency Response Times:

- In towns like Tiruchengode, response times may be quicker (around 15-30 minutes) due to closer proximity to emergency services. Rural stretches may experience delays, extending response times potentially to 45 minutes or more.

11. Overall Summary of Risk Assessment:

- The route presents moderate risks primarily around congestion and during adverse weather conditions. Hazards could include variable road quality, limited shoulder space, and traffic density in semi-urban areas. Vigilant driving, adherence to local regulations, and preparedness for weather conditions are essential for a safe journey.

By considering the highlighted factors and maintaining cautious driving practices, many potential hazards can be mitigated along this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43961, 77.87341	30 KM/Hr
3	Turn	Medium	11.43968, 77.87345	30 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	High	11.38346, 77.89975	15 KM/Hr
6	Turn	High	11.38190, 77.90254	15 KM/Hr
7	Turn	Medium	11.38182, 77.90259	30 KM/Hr
8	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
9	Turn	High	11.27732, 78.08017	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
8	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
7	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38346, 77.89975



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38190, 77.90254



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38182, 77.90259



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.37841, 77.90140



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.27732, 78.08017
